

Consider a litter of rabbits in which the foundation of the father's fur is black, "Bb" and the mother of the litters also has fur with a black foundation, "Bb."

- Complete the Punnett Square to show the possible traits of a litter of rabbits if their father has black fur (Bb) and their mother has black fur (Bb) as well.

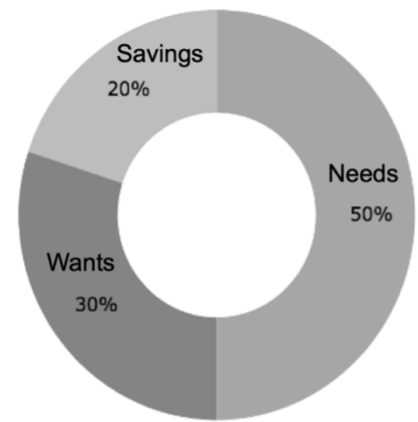
Punnett Square		Father's Code	
		B	b
Mother's Code	B	BB	Bb
	b	Bb	bb

- What is the likelihood that these two parents will have offspring with brown fur (bb)?
 $\frac{1}{4}$
- What is the likelihood that these two parents will have offspring with black fur (BB and Bb)?
 $\frac{3}{4}$
- Suppose these parents have 25 kits, or baby rabbits, with brown (bb) in one year. Predict the total amount of kits these parents had.
 $\frac{1}{4} \cdot x = 25 \rightarrow x = 100 \text{ kits}$
- Predict the number of kits these parents had that had black fur that same year.

$$\frac{3}{4} \cdot 100 = 75 \text{ kits}$$

Savannah is very careful about what she does with the money she makes from her part time job. The graph shows the probability of how the money she makes is used.

Probability Distribution of Income



6. If Savannah has a paycheck of \$200, predict how much money is being put into savings.

$$200 \cdot 0.20 = \$40$$

7. Suppose that she spends \$36 on things she wants but doesn't need. Predict the total amount of money she had that paycheck.

$$0.30x = \$36 \rightarrow x = \$120$$

8. In one year, Savannah made \$5,000. Estimate how much of that money she spent on things she needed.

$$5,000 \cdot 0.5 = \$2,500$$

9. Tyrone owns 84 movies. One fourth of those are horror movies. If Tyrone randomly picks 8 movies without looking, predict how many of those movies are *not* horror.

$$\frac{3}{4} \text{ are not horror} \quad \frac{3}{4} \cdot 8 = 6 \text{ movies are not horror}$$

10. The Taco Bell needs to restock their drinks. Based off on past sales, they determine that $\frac{2}{3}$ of their order needs to be for cola products. After restocking, 15 random customers come in to order and get drinks with their meals. Predict how many of those customers did *not* order a cola product.

$$\frac{1}{3} \cdot 15 = 5 \text{ customers did not order a cola product}$$